VEO Patent Strengthening Instructions

Non-Provisional Upgrade — Patent #3

Method and System for Generating, Scoring, Signing, Anchoring, and Verifying Verifiable Entropy Objects

Date: June 25, 2026

> **Scope: Expand protection from a specific implementation into a foundational infrastructure layer.**
>
> **Rule: Do NOT rewrite the existing invention. Preserve all existing content and ADD the following concepts, embodiments, and claim coverage.**

1. Reframe VEO as a Universal Uncertainty Artifact

Current Framing

Verifiable Entropy Object is a structured entropy object.

Expand To

A Verifiable Entropy Object (VEO) is a **portable, machine-verifiable uncertainty artifact** that encapsulates one or more entropy sources, provenance information, quality assessments, cryptographic attestations, trust metadata, and verification evidence.

Add Language That

VEO is analogous to:

- **PDF** for documents
- **JWT** for identity
- **VEO** for uncertainty

> **Goal:** Protect the concept of uncertainty becoming a transferable, auditable digital artifact — not merely a random number.

2. Add AI Decision Attestation Embodiments

The provisional mentions AI in the background but does not claim it.

Add Embodiments Covering

- LLM tool selection
- Agent routing
- Multi-agent arbitration
- Resource allocation
- Autonomous task assignment
- Sampling decisions
- Autonomous execution approval
- Model selection
- AI governance workflows

Example Embodiment

An autonomous software agent generates or references a VEO prior to making a non-deterministic decision and records the VEO identifier alongside the decision outcome.

Example Claim Language

A computer-implemented method comprising: generating a verifiable entropy object; associating the verifiable entropy object with an autonomous agent decision; recording a decision outcome linked to the verifiable entropy object; enabling independent verification of the randomness influencing the decision.

> **Goal:** Directly cover future AI infrastructure.

3. Introduce Decision Receipt / Decision Attestation Concepts

New Embodiment: Decision Receipt

```
{
  "decision_id": "...",
  "decision_type": "tool_selection",
  "selected_option": "Tool B",
  "confidence": 0.81,
  "veo_id": "...",
  "signature": "...",
  "anchor": "..."
}
```

Describe

- Human decisions
- AI decisions
- Multi-agent decisions
- Governance votes
- Automated execution events

Possible Terminology

- Decision Attestation Record
- Decision Receipt
- Verifiable Decision Record
- Decision Evidence Object

> **Goal:** Create continuation opportunities around auditable decision systems.

4. Generalize Signature Systems

Current filing references: Ethereum, EIP-191, secp256k1.

Expand Claims and Specification to Cover

- secp256k1
- Ed25519
- RSA
- Post-quantum signatures
- Future asymmetric signature systems

Use Language Such As

> *Any asymmetric cryptographic signature scheme capable of proving authenticity and permitting verification.*

Avoid limiting claims to Ethereum.

> **Goal:** Prevent easy design-around.

5. Generalize Hash Functions

Current examples focus on: SHA-256.

Expand Specification to Cover

- SHA-256, SHA-3, Keccak, Blake2, Blake3
- Future cryptographic hash functions

Use Language Such As

> *One or more cryptographic hash functions.*

> **Goal:** Future-proof claim scope.

6. Generalize Anchoring Targets

Current filing primarily assumes: blockchain anchoring.

Expand To

- Public blockchains
- Consortium chains
- Private chains
- Append-only ledgers
- Transparency logs
- Notarization networks
- Distributed timestamping systems

Claim

> *"Immutable verification substrate" instead of only "blockchain."*

> **Goal:** Protect the trust model rather than a specific ledger.

7. Expand Verification Framework into Trust State Machine

Current verification levels: Structurally Valid, Cryptographically Verified, Anchored Verified, Policy Failed, Invalid.

Describe This As

A **trust state machine** where an object transitions between verification states.

Possible Additional States

- pending
- partially verified
- expired
- revoked
- superseded

Potential Claim

> *Assigning an object to one of a plurality of trust states based on progressively evaluated verification criteria.*

> **Goal:** Separate patentable value from the entropy itself.

8. Expand ECS Beyond Fixed Weights

Current ECS: Fixed dimensions with fixed weights.

Add Embodiments Covering

- Dynamic weighting
- Policy-based weighting
- Industry-specific weighting
- Machine-learned weighting
- Regulator-defined weighting

Language

> *The dimensions and weighting coefficients may be modified based on application requirements.*

> **Goal:** Avoid limiting protection to current percentages.

9. Add Interoperability Claims

Add Embodiments Where

- One provider generates a VEO
- Another provider verifies it
- Another provider consumes it

Describe

- Vendor-neutral verification
- Cross-platform entropy portability
- Third-party auditability

Key Statement

> *A verifier need not trust the original generator.*

> **Goal:** Move toward standards-layer positioning.

10. Add Entropy-as-a-Service Embodiments

Add Service Architecture

- Entropy generation service
- VEO issuance service
- VEO verification service
- ECS scoring service
- Anchoring service

Potential Claim

> *Issuing verifiable entropy objects through a network-accessible service.*

> **Goal:** Protect future API businesses.

11. Add Multi-Party Governance Embodiments

Examples

- DAO voting
- Lottery systems
- Regulated gaming
- Compliance approvals
- Corporate governance

Where: VEO is referenced during outcome generation; outcome remains independently auditable.

> **Goal:** Increase licensing opportunities.

12. Add AI Audit Trail Language

New Section: AI Auditability

Describe

- Decision provenance
- Replayability
- Regulatory review
- Model governance
- Post-event investigation

Example

> *A third party may reconstruct the decision context by verifying the associated VEO and related decision records.*

> **Goal:** Position VEO as future AI governance infrastructure.

Strategic Objective

The non-provisional should no longer read as:

> *"A better random number system."*

It should read as:

> *"A universal framework for packaging, attesting, transporting, verifying, and auditing uncertainty and non-deterministic decisions across software systems, AI agents, governance systems, gaming systems, and distributed networks."*

That broader framing maximizes:

- **Long-term patent value**
- **Continuation opportunities**
- **Licensing potential**
- **Acquisition attractiveness**

Reference Documents

#	Document	Purpose
1	Patent3_VEO_Provisional.pdf	Filed provisional application

2	RFC-0001-VEO1.md	Frozen protocol specification
3	ECS-v1.md	Frozen scoring specification
4	OpenRNG_Whitepaper_v1.md	Category-defining whitepaper
5	Agent Arbiter demo	Working AI decision attestation example